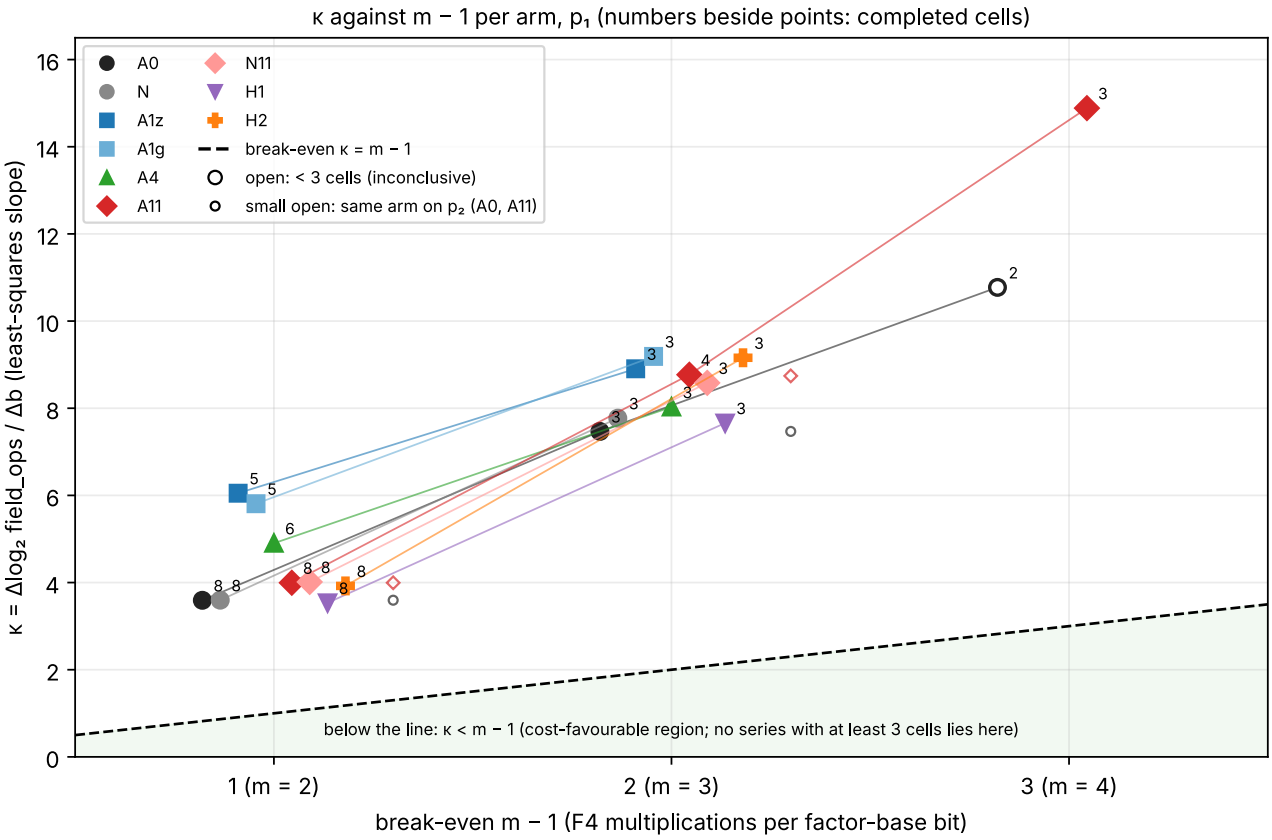
Factor-base designs: close-out of the prime-field slope line (review-20261019T1632Z)

Kind: review of committed results; no new run. Observations only. Nothing here prices a pipeline, reports S or shows a speedup. Every number is copied from `runs/analysis-amendment4.txt` (Amendment 4, run-20261016T0616Z), and the graph is parsed from that file by `plot_kappa.py` with no further computation. The protocol is `PREREGISTRATION.md` Amendments 4 and 5.

Requested scope and what was done

- **Close-out statuses** for every measured prime-field arm: README §2 (table), with κ and the cell count copied from the analysis file.
- **Amendment 5** appended to `PREREGISTRATION.md`: κ is the primary score for later rounds, with the m − 1 break-even and Amendment 4′s thresholds; σ is secondary. It changes no existing verdict.
- **Owed items:** the novelty audit (`NOVELTY_AUDIT.md`, abstracts and snippets only, no full text read); the Mathlib bridges (not attempted: the cache did not fit the disk before, and it is shared); A0′s third m = 4 cell (not run: needs more than 13 GB of memory or isolab credentials, which this container lacks).

κ against the break-even, per arm



Source: runs/analysis-amendment4.txt (Amendment 4, run-20261016T0616Z). Points at one m − 1 are offset horizontally for legibility only.

Figure 1: κ = $\Delta\log_2$ field_ops / Δb (least-squares slope over completed cells) against the break-even m − 1, one marker per arm and arity on p_1. Numbers beside markers are completed cells. Open markers have fewer than 3 cells. Small open markers are A0 and A11 on p_2 (F1). The dashed line is κ = m − 1. Points at one m − 1 are offset horizontally for legibility. Source `runs/analysis-amendment4.txt`, run run-20261016T0616Z, `plot_kappa.py`.

arm	m = 2: κ (cells)	m = 3: κ (cells)	m = 4: κ (cells)	κ/(m − 1), m = 2	m = 3	status
A0	3.593 (8)	7.468 (3)	10.773 (2)	3.59	3.73	unfavourable at m = 2, 3; m = 4 inconclusive
N	3.596 (8)	7.772 (3)	not run	3.60	3.89	unfavourable at m = 2, 3
A1z	6.050 (5)	8.905 (3)	not run	6.05	4.45	unfavourable at m = 2, 3
A1g	5.809 (5)	9.189 (3)	not run	5.81	4.59	unfavourable at m = 2, 3
A4	4.902 (6)	8.028 (3)	not run	4.90	4.01	unfavourable at m = 2, 3
A11	3.995 (8)	8.769 (4)	14.886 (3)	3.995	4.38	unfavourable at m = 2, 3, 4

N11	4.014 (8)	8.582 (3)	not run	4.01	4.29	unfavourable at m = 2, 3
H1	3.535 (8)	7.660 (3)	not run	3.54	3.83	unfavourable at m = 2, 3
H2	3.923 (8)	9.158 (3)	not run	3.92	4.58	unfavourable at m = 2, 3

The κ/(m − 1) columns are quotients of the two committed numbers (derived, each recomputed here). At m = 4: A0 10.773 / 3 = 3.59 (2 cells, inconclusive) and A11 14.886 / 3 = 4.96 (3 cells).

Observations:

- Every series with at least 3 completed cells has $\kappa \geq m - 1$ (23 series; Amendment 4′s 22 at m = 2, 3, plus A11 at m = 4). The smallest κ/(m − 1) among them is 3.54 (H1, m = 2) and the largest is 6.05 (A1z, m = 2).
- A0: κ/(m − 1) = 3.59, 3.73, 3.59 at m = 2, 3, 4 on 8, 3 and 2 cells. A11: 3.995, 4.38, 4.96 on 8, 4 and 3 cells. This is an observation on few cells, one prime for the m = 4 points, with no interval recorded. **Nothing is extrapolated** to m > 4, larger t or deployed sizes.
- The p_2 κ for A0 and A11 (3.595 and 3.995 at m = 2; 7.468 and 8.742 at m = 3) is within 0.03 of p_1.

Why κ replaces σ as the primary score (Amendment 5)

- A1z and A1g sit at D = 8 in every cell (σ = 0.0000, **candidate**), yet their κ (6.050, 5.809 at m = 2; 8.905, 9.189 at m = 3) is above A0′s (3.593; 7.468). At m = 3, t = 4 their max_cols are 39 648 and 41 629 against A0′s 12 314.
- A11 has σ = 0.1786 at m = 2 and 0.3000 at m = 3 on p_1 (0.0000 on p_2), yet κ = 3.995, 8.769 and 14.886 at m = 2, 3, 4, above A0′s at m = 2 and 3.

Limits

- A0 at m = 4 has 2 completed cells; its t = 4 cell exhausts the 13 GB memory limit (C2, 512.8 s and 536.9 s). The third cell is owed and blocked.
- Only A0 and A11 were run at m = 4, so no other arm has a status at m = 4.
- Verdicts are read at the sizes measured. The Lean break-even is a model statement. The Mathlib bridges behind K7 and K8 remain owed.
- `NOVELTY_AUDIT.md` rests on abstracts, slides and search snippets. Statuses: A11, A4, H1/H2 partly prior; A1 not found in the sources checked.

Records

`runs/analysis-amendment4.txt`, `runs/C1-rescore.jsonl`, `runs/C2-*`, `runs/C4-*`, run report `reports/run-20261016T0616Z/REPORT.pdf`. Graph checked and left unchanged: the run report's `ops_vs_b.svg` and `ablation.svg` (no new data). Commit of this review: see the git history of this directory.